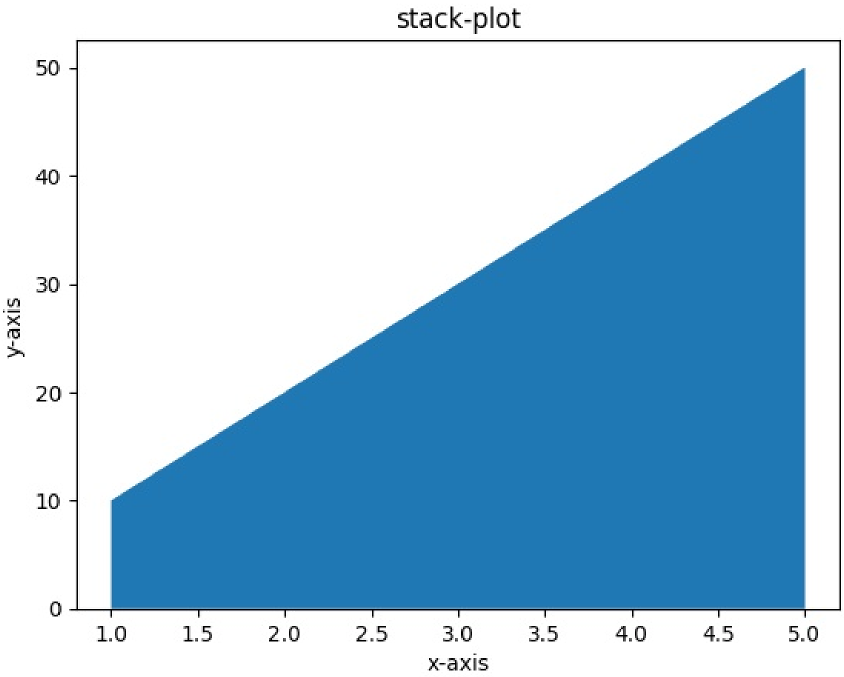
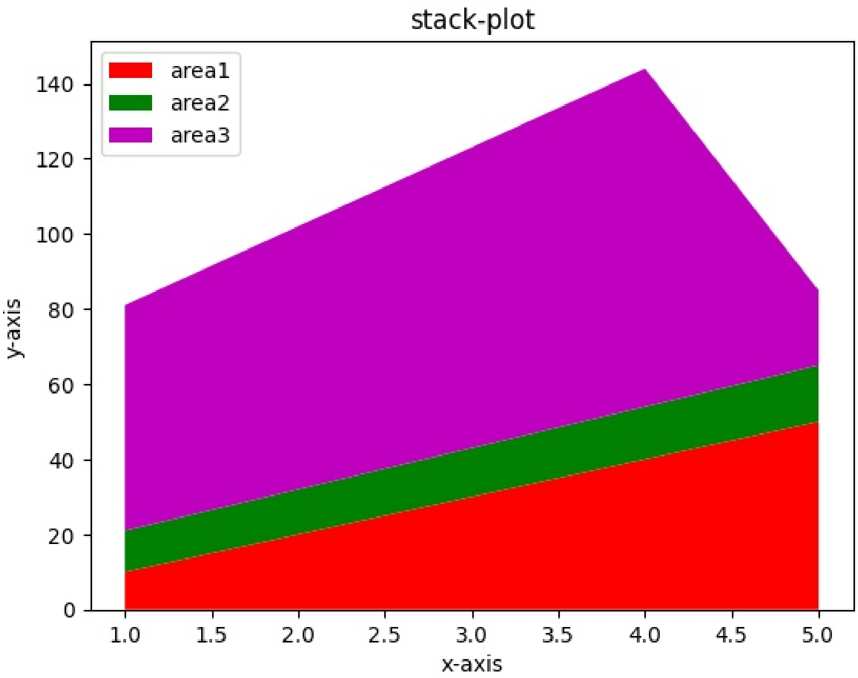
Stack-plot

In [3]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
area=[10,20,30,40,50]


plt.title("stack-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


plt.stackplot(x,area)
plt.show()
```



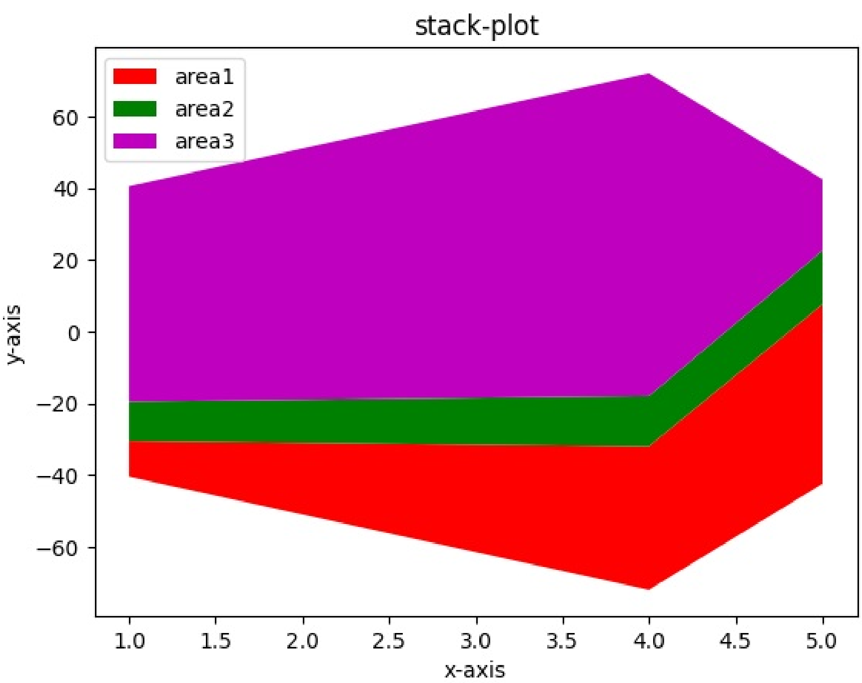
multiple Area chart

In [7]:
```
import matplotlib.pyplot as plt

x=[1,2,3,4,5]


area1=[10,20,30,40,50]
area2=[11,12,13,14,15]
area3=[60,70,80,90,20]

lab=["area1","area2","area3"]
c=["r","g","m"]

plt.title("stack-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


plt.stackplot(x,area1,area2,area3,labels=lab,colors=c)
plt.legend(loc="upper left")
plt.show()
```



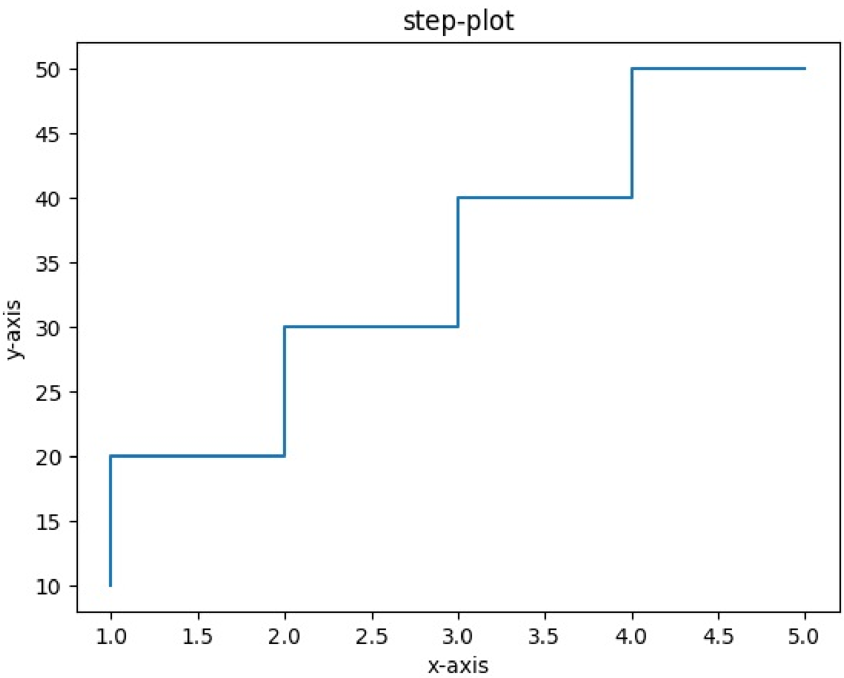
baseline="sym","zero","wiggle"

In [8]:
```
import matplotlib.pyplot as plt

x=[1,2,3,4,5]


area1=[10,20,30,40,50]
area2=[11,12,13,14,15]
area3=[60,70,80,90,20]

lab=["area1","area2","area3"]
c=["r","g","m"]

plt.title("stack-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


plt.stackplot(x,area1,area2,area3,labels=lab,colors=c,baseline="sym")
#  ,baseline="sym","zero","wiggle"
plt.legend(loc="upper left")
plt.show()
```



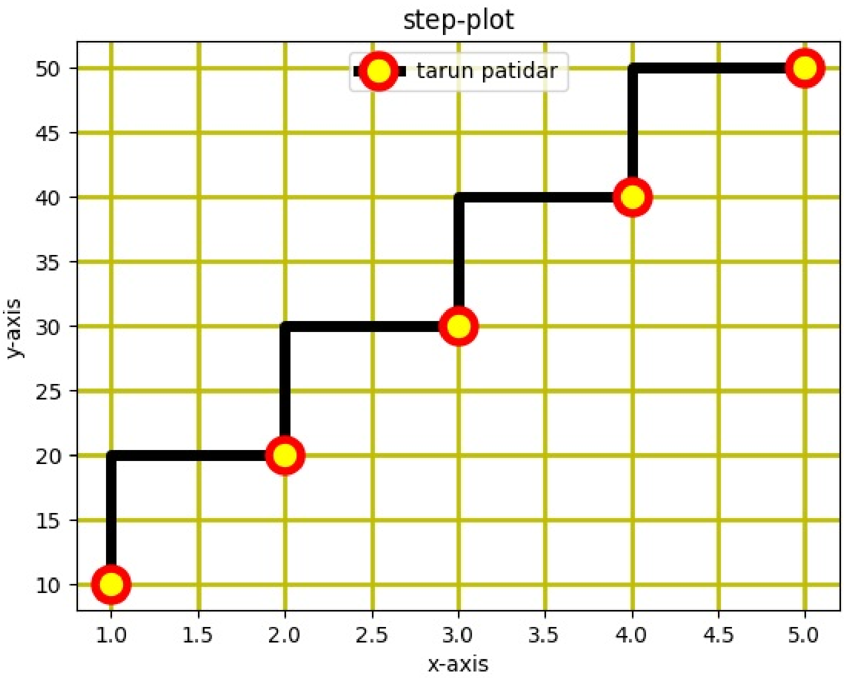
step-plot

In [10]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,30,40,50]
plt.title("step-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.step(x,y)
plt.show()
```



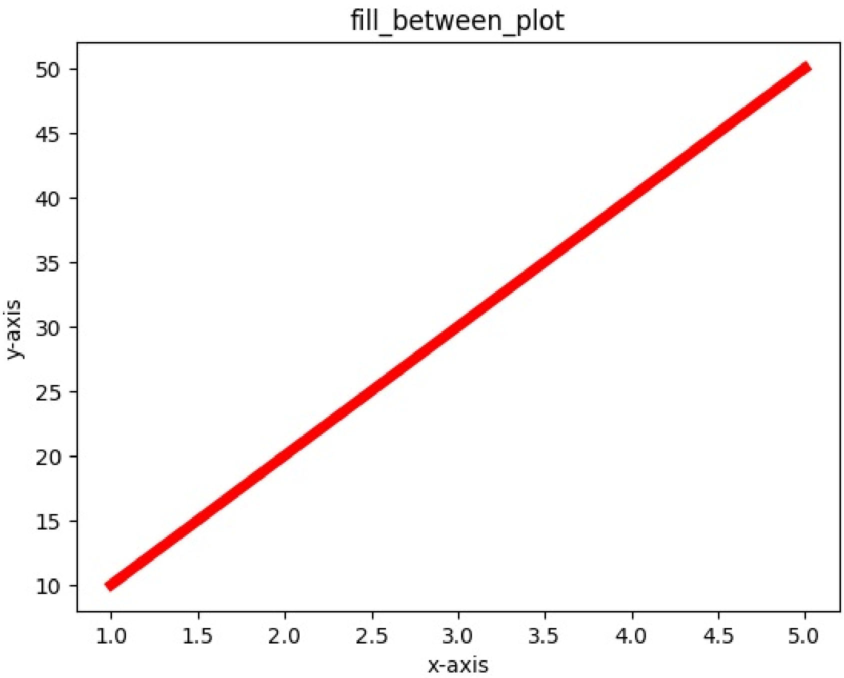
marker parameter

In [29]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,30,40,50]
plt.title("step-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")

plt.step(x,y,color="k",linewidth=5, marker="o", markersize=15,markeredgewidth=4,

    markerfacecolor="yellow",markeredgecolor="red",label="tarun patidar")

plt.grid(linewidth=2,color="y")

plt.legend(loc="upper center")

plt.show()
```



Linear Plot

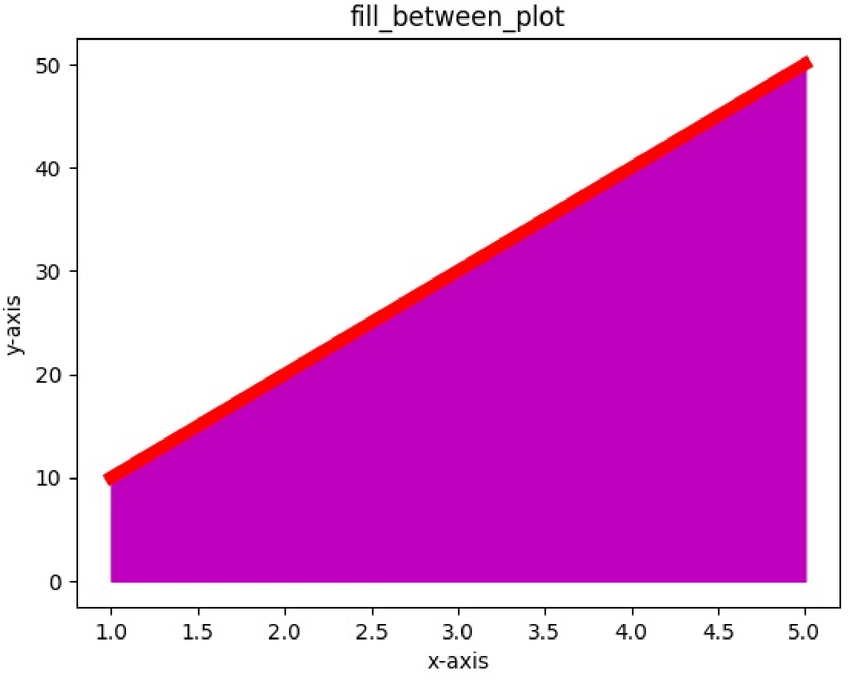
In [34]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,30,40,50]

plt.title("fill_between_plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")

plt.plot(x,y,color="r",linewidth=5)
plt.show()
```



plt.fill_between(x,y,color="m")

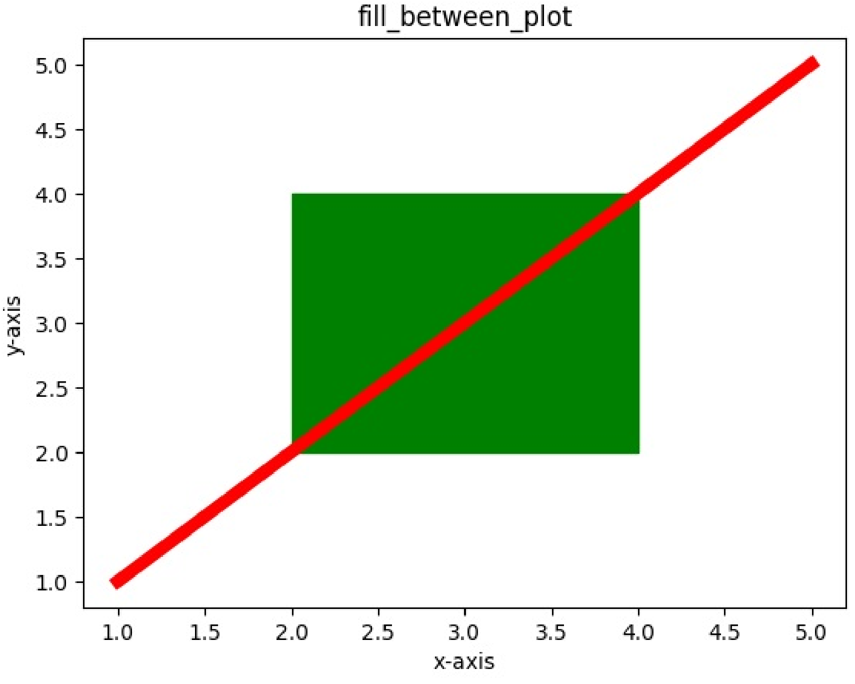
In [40]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,30,40,50]

plt.title("fill_between_plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")
```

```
plt.fill_between(x,y,color="m")

plt.plot(x,y,color="r",linewidth=6)
plt.show()
```



plt.fill_between(x=[2,4],y1=2,y2=4,color="g")

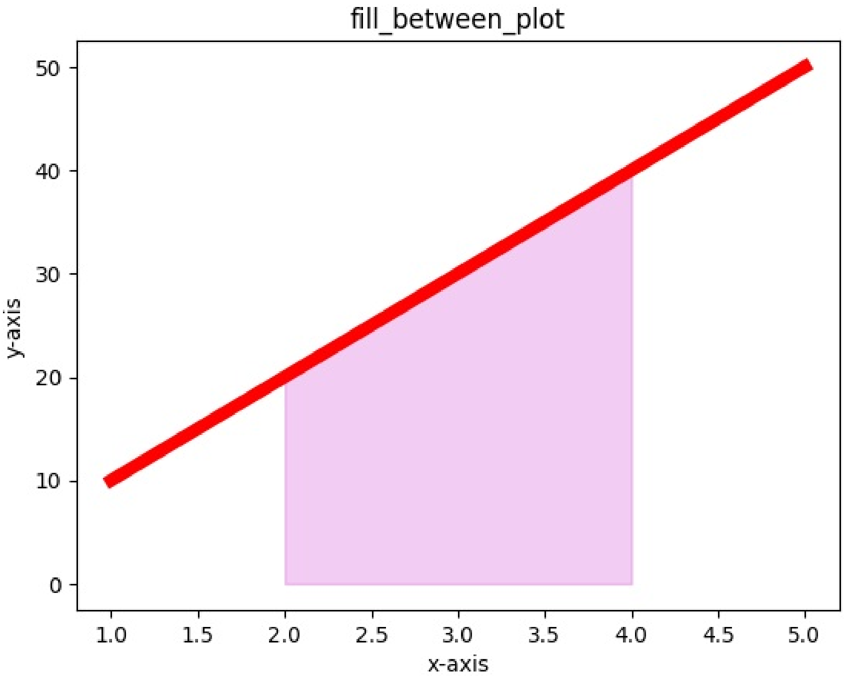
In [39]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[1,2,3,4,5]

plt.title("fill_between_plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


plt.fill_between(x=[2,4],y1=2,y2=4,color="g")

plt.plot(x,y,color="r",linewidth=6)
plt.show()
```



where paremeter

In [44]:
```
import matplotlib.pyplot as plt
```

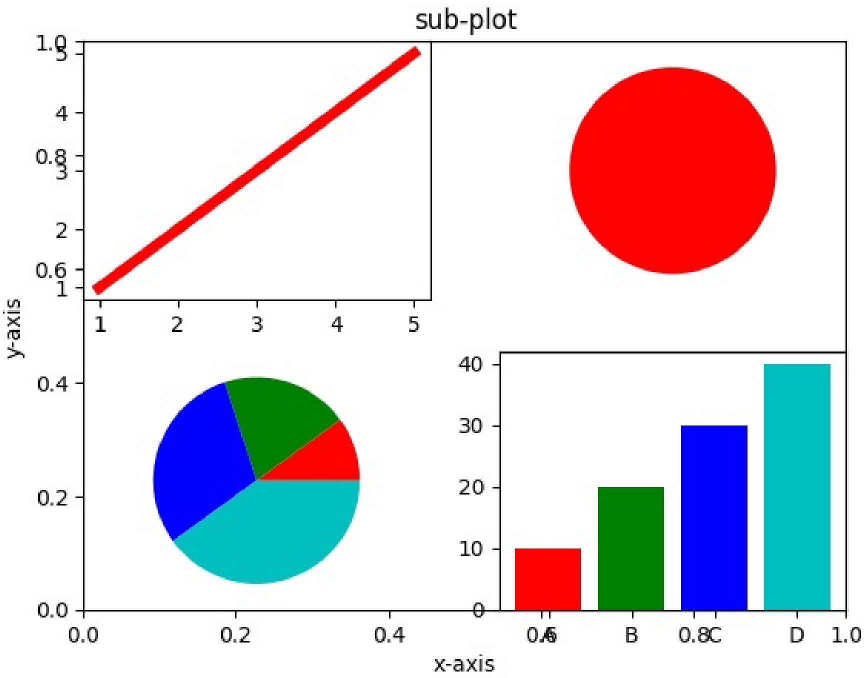
```
import numpy as np

x=np.array([1,2,3,4,5])
y=np.array([10,20,30,40,50])

plt.title("fill_between_plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


plt.fill_between(x,y,color="m",where=(x>=2)&(x<=4),alpha=0.2)

plt.plot(x,y,color="r",linewidth=6)
plt.show()
```



sub-plot

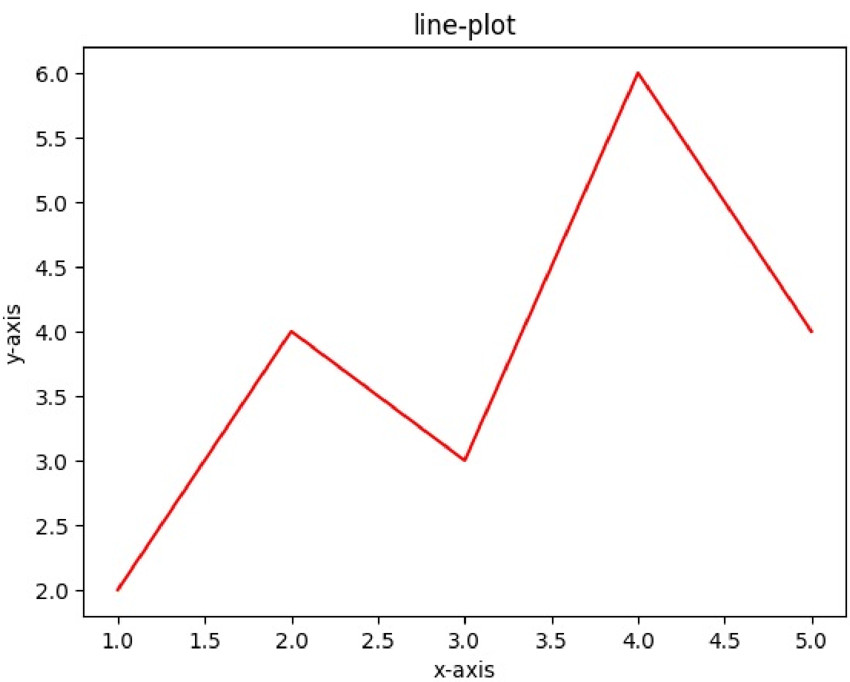
In [59]:

```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[1,2,3,4,5]
c=["r","g","b","c"]

plt.title("sub-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


#simple plot chart
plt.subplot(2,2,1)
plt.plot(x,y,color="r",linewidth=5)


# dot pie chart
plt.subplot(2,2,2)
plt.pie([1],colors=c)


#pie chart
x2=[10,20,30,40]
plt.subplot(2,2,3)
plt.pie(x2,colors=c)




#Bar chart
x3=["A","B","C","D"]
y3=[10,20,30,40]
plt.subplot(2,2,4)
plt.bar(x3,y3,color=c)

plt.show()
```



plt.savefig("line",dpi=2000,facecolor="g") #dot-par-inch

In [71]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[2,4,3,6,4]


plt.title("line-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


plt.plot(x,y,color="r")

plt.savefig("line",dpi=2000,facecolor="g",transparent=True,bbox_inches="tight")    #dot-par-inch

plt.show()
```

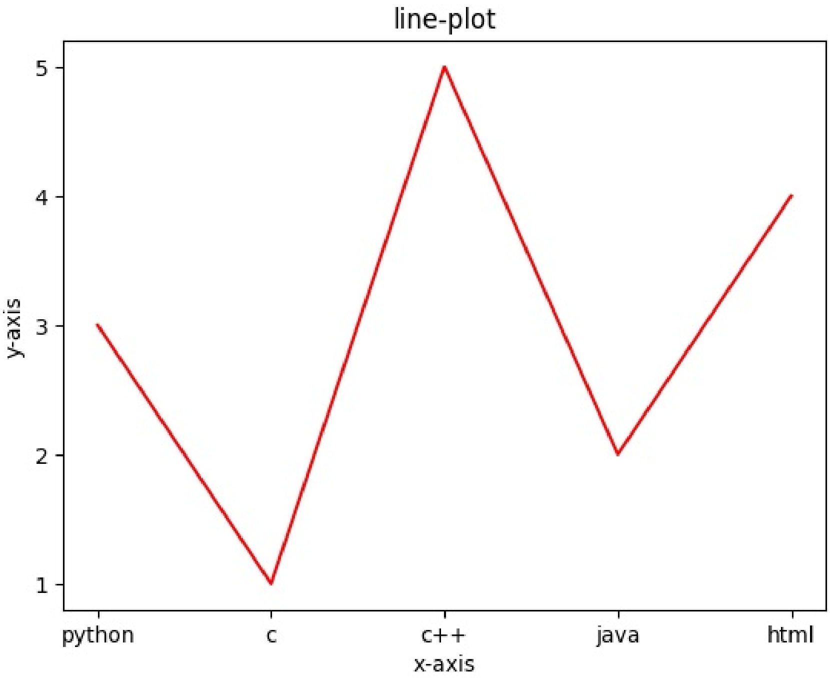


plt.xticks(x,labels=lab1)

In [76]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
```

```
y=[3,1,5,2,4]


plt.title("line-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")

lab1=["python","c","c++","java","html"]
plt.xticks(x,labels=lab1)
plt.yticks(x)

plt.plot(x,y,color="r")
```

Out[76]: [<matplotlib.lines.Line2D at 0x2023a5c9d30>]

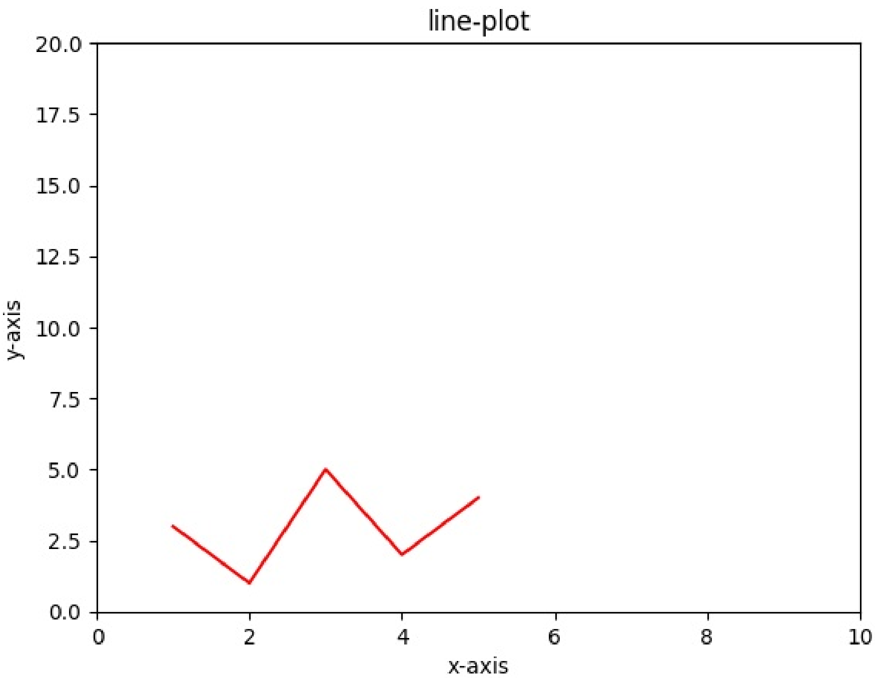


axes limit()

In [79]:

```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[3,1,5,2,4]


plt.title("line-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


plt.xlim(0,10)
plt.ylim(0,20)





plt.plot(x,y,color="r")
```

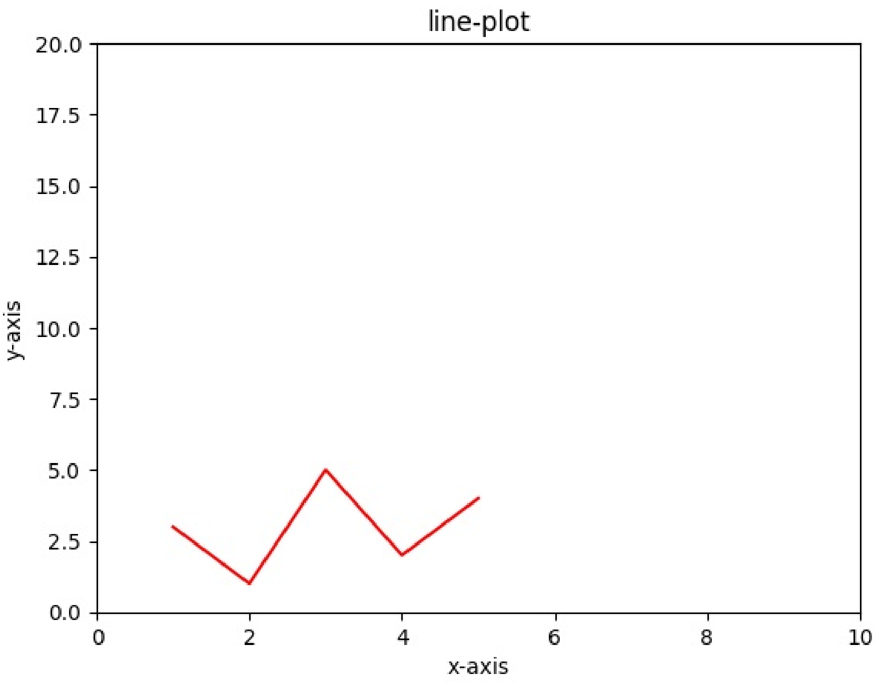
Out[79]: [<matplotlib.lines.Line2D at 0x2023a6b9400>]



plt.axis([0,10,0,20]) # (0 to 10) x-axis and (0 to 20) y-axis

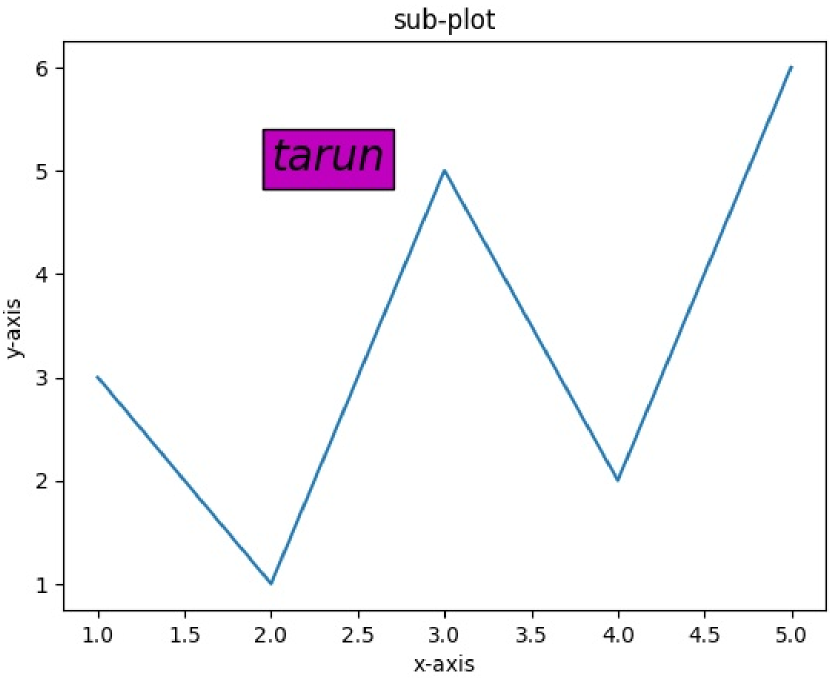
In [80]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[3,1,5,2,4]


plt.title("line-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")


# plt.xlim(0,10)
# plt.ylim(0,20)

plt.axis([0,10,0,20])  # (0 to 10)  x-axis and (0  to 20) y-axis




plt.plot(x,y,color="r")
```

Out[80]:
```
[<matplotlib.lines.Line2D at 0x2023a4ad160>]
```



plt.text(2,5,"tarun",fontsize=20,style="italic",bbox={"facecolor":"m"})

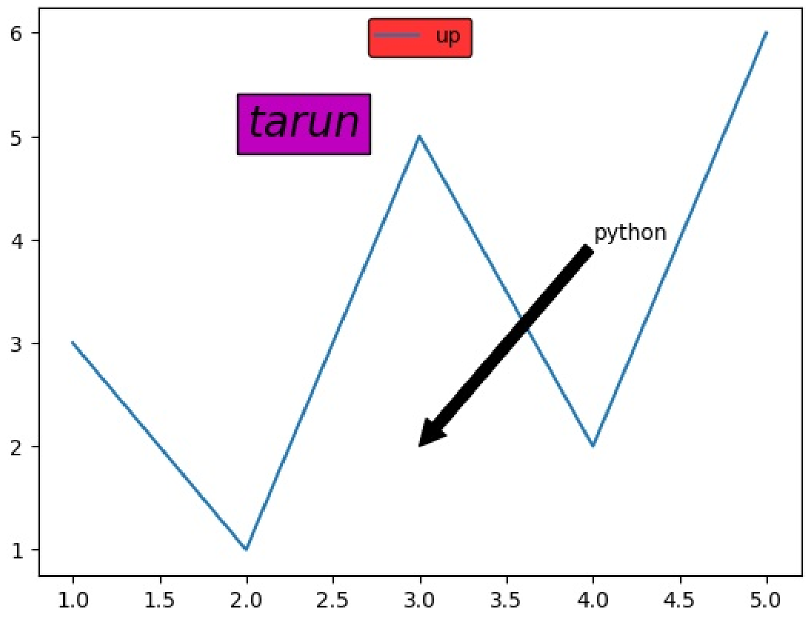
In [87]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[3,1,5,2,6]


plt.title("sub-plot")
plt.xlabel("x-axis")
plt.ylabel("y-axis")

plt.text(2,5,"tarun",fontsize=20,style="italic",bbox={"facecolor":"m"})

# b_box => (key:value)


plt.plot(x,y)
plt.show()
```



plt.annotate()

In [111…

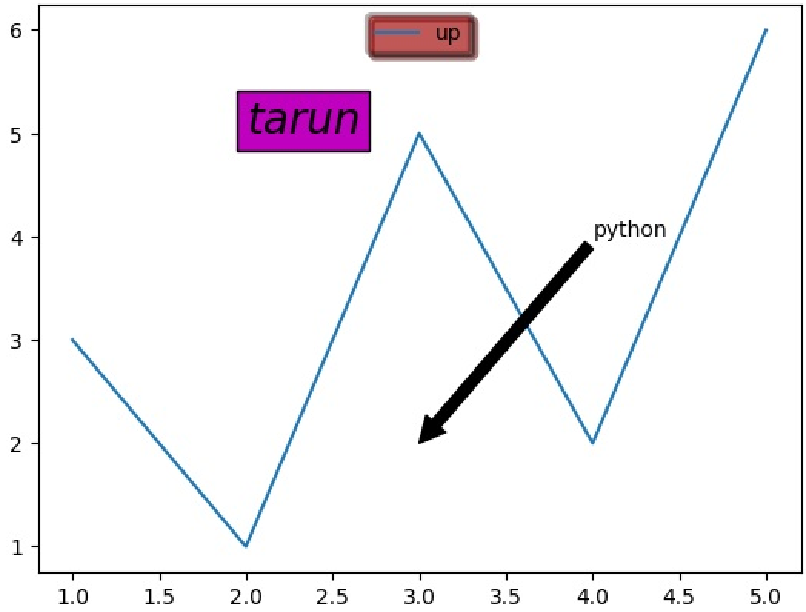
```
import matplotlib.pyplot as plt

x=[1,2,3,4,5]
y=[3,1,5,2,6]

plt.plot(x,y)  # □□□□□

plt.text(
    2,5,
    "tarun",
    fontsize=20,
    style="italic",
    bbox={"facecolor":"m"}
)

plt.annotate(
    "python",
    xy=(3,2),
    xytext=(4,4),
    arrowprops=dict(
        facecolor="black",
        width=5

    )
)


plt.legend(["up"],loc="upper center",facecolor="r",edgecolor="black")

plt.show()
```



leg.get_frame().set_linewidth(3)

In [116…

```
import matplotlib.pyplot as plt

x=[1,2,3,4,5]
y=[3,1,5,2,6]

plt.plot(x,y)  # □□□□□

plt.text(
    2,5,
    "tarun",
    fontsize=20,
    style="italic",
    bbox={"facecolor":"m"}
)

plt.annotate(
    "python",
    xy=(3,2),
    xytext=(4,4),
    arrowprops=dict(
        facecolor="black",
        width=5

    )
)


leg = plt.legend(["up"], loc="upper center",
                 facecolor="r", edgecolor="black",framealpha=0.3,shadow=True)

leg.get_frame().set_linewidth(3)

plt.show()
```


up
tarun
python
6
5
4
3
2
1
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0